



HSB

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Project Proposal

Modeling Biological Pattern Formation via Gray-Scott
Reaction-Diffusion Systems

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June 24, 2026

1 Abstract

This project explores the mathematical modeling and numerical simulation of biological pattern formation using partial differential equations (PDEs). Specifically, we investigate the Gray-Scott model of reaction-diffusion systems, which describes the chemical interactions and spatial dispersion of two virtual substances. Operating on the principles of Alan Turing's morphogenetic theory, this system spontaneously generates complex, organic structures such as spots, stripes, and moving waves from nearly homogeneous initial states.

Using Python, this project implements a finite-difference time-domain (FDTD) scheme to discretize and solve the governing PDEs. The final deliverable features a dynamic, real-time 2D visualization environment allowing live parameter manipulation and spatial perturbations via mouse interaction. This framework serves to evaluate system stability, bifurcations, and the structural sensitivity of natural pattern-generation algorithms.

2 Motivation, Application Scenario, Requirements

In 1952, Alan Turing demonstrated that a system of reacting and diffusing chemicals could break spatial symmetry, giving rise to stable, macroscopic patterns. This counterintuitive concept, where diffusion, typically a homogenizing process, drives structural chaos and localized order, is essential to understanding developmental biology and self-organization in non-equilibrium systems.

This project offers three distinct tiers of application scenarios:

- **Developmental Zoology:** The simulation of skin, fur or scale patterns. Also the process of Morphogenesis, which describes the process that causes an organism, tissue, or cell to develop its shape.
- **Bio-Medical Engineering:** The analysis of geometric structures formed by growing colonies of bacteria under stress, which follow reaction-diffusion laws.
- **Procedural Asset Generation:** A highly practical, modern software engineering use case. In modern game engines, storing high resolution textures takes up immense memory. A solution, instead of saving textures, could consist of using a reaction-diffusion script. The computer then generates highly complex, organic textures procedurally. This allows for a huge amount of saved storage and for infinitely detailed zoom-ins.

2.1 System Requirements

1. **Mathematical Fidelity:** Accurate numerical estimation of spatial derivatives via localized stencil operations.

2. **Computational Efficiency:** Real-time array processing to handle rapid iterative updates without frame drops.
3. **User Interaction:** Ability to actively inject virtual chemicals or alter global reaction kinetics on the fly to observe resilience and structural adaptability.

3 Related Work

- **Turing, A. M. (1952):** *The Chemical Basis of Morphogenesis*. This foundational paper mathematically established that a reaction-diffusion system can generate self-organizing spatial patterns.
- **Gray, P., & Scott, S. K. (1983):** *Autocatalytic reactions in the isothermal, continuous stirred tank reactor*. They simplified the chemical equations into a highly dynamic cubic autocatalytic model.
- **Pearson, J. E. (1993):** *Complex Patterns in a Simple System*. Pearson mapped out the vast parametric landscape (F, k) of the Gray-Scott model, categorizing distinct morphological regimes (chaos, solitons, mitosis).

4 Development Target (Implementation and Examples)

4.1 Mathematical Foundation

The Gray-Scott model describes the interaction of two chemical concentrations, $u(\vec{x}, t)$ and $v(\vec{x}, t)$, governed by the following coupled nonlinear system of partial differential equations:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + F(1 - u) \quad (1)$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (F + k)v \quad (2)$$

Where:

- D_u, D_v : Diffusion coefficients of the respective substances.
- ∇^2 : The Laplacian operator representing spatial diffusion $(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})$.
- uv^2 : The non-linear, autocatalytic reaction term (one molecule of U combined with two of V creates more V).
- F : The feed rate (replenishing U and draining the system).
- k : The kill rate (removing V from the system).

4.2 Numerical Discretization (The Finite-Difference Method)

To solve this in Python on a discrete 2D grid with spacing $\Delta x, \Delta y$ and time-step Δt , we approximate the continuous derivatives. The spatial Laplacian ∇^2 at grid coordinate (i, j) is approximated using a 5-point discrete stencil:

$$\nabla^2 q_{i,j} \approx \frac{q_{i+1,j} + q_{i-1,j} + q_{i,j+1} + q_{i,j-1} - 4q_{i,j}}{(\Delta x)^2} \quad (1)$$

The temporal update mimics the explicit Forward-Euler integration method established in the earlier coursework:

$$u_{i,j}^{n+1} = u_{i,j}^n + \Delta t (D_u \nabla^2 u_{i,j}^n - u_{i,j}^n (v_{i,j}^n)^2 + F(1 - u_{i,j}^n)) \quad (2)$$

$$v_{i,j}^{n+1} = v_{i,j}^n + \Delta t (D_v \nabla^2 v_{i,j}^n + u_{i,j}^n (v_{i,j}^n)^2 - (F + k)v_{i,j}^n) \quad (3)$$

5 Use of Tools

- **Language:** Python 3.x
- **Numerical Core (numpy):** Matrix operations and vectorization. Vectorized slicing (`U[2:, 1:-1] + U[:-2, 1:-1]...`) will replace nested `for`-loops to achieve maximum computational performance.
- **Visualization (matplotlib.pyplot):** `imshow()` map rendering using high-contrast scientific colormaps (`'magma'`, `'viridis'`).
- **Animation Engines (matplotlib.animation.FuncAnimation):** For real-time, frame-by-frame updates of the numerical solution.
- **UI Controls (matplotlib.widgets):** Implementation of interactive sliders for F and k , alongside `canvas.mpl_connect('button_press_event')` to facilitate custom, localized chemical injection via mouse clicks.

6 Evaluation Methods and Expected Results

- **Parametric Exploration:** Systematically sampling the parameter space across a grid of $(F, k) \in [0.01, 0.09] \times [0.04, 0.07]$.
- **Stability Limits:** Verifying the Courant-Friedrichs-Lewy (CFL) numerical stability criterion for explicit diffusion methods to avoid numerical explosion:

$$\Delta t \leq \frac{(\Delta x)^2}{4 \max(D_u, D_v)} \quad (1)$$

6.1 Expected Results

- **Pattern Stratification:** Generation of a comprehensive static multi-panel chart (Dashboard Grid) documenting macrostructures corresponding to Pearson’s classification.
- **Dynamic Behavior:** Successful instantiation of an animated interactive interface. The user will observe phase shifts (e.g., active finger-print lines splitting into independent spots) when dynamically dragging parameter sliders or clicking the canvas to induce a state perturbation.

7 Work Schedule/Load

The project will be executed over a five-week timeline to meet all academic milestones. All team members will work on all aspects of this project.

- **Week 1: Mathematical Foundations & Scripting Architecture**
Solidifying the mathematical model, designing grid boundaries, and setting up the vectorized discrete Laplacian stencil in NumPy.
- **Week 2: Numerical Solver Implementation**
Coding the main simulation time-loop and calibrating time-steps (Δt) against diffusion limits to prevent system instabilities.
- **Week 3: Animation Rendering & Interactive Widgets**
Integrating `FuncAnimation` for smooth UI rendering and building interactive widgets (parameter sliders, mouse-click event-handlers).
- **Week 4: Parameter Space Mapping & Analysis**
Running parallel batch simulations to isolate specific morphological patterns (spots, stripes, solitons) and saving specific states for documentation.
- **Week 5: Code Optimization, Review & Final Report**
Refactoring bottleneck operations, conducting performance metrics, and compiling the comprehensive project write-up.

References

- [1] Turing, A. M. (1952). *The chemical basis of morphogenesis*. Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences, 237(641), 37-72.
- [2] Gray, P., & Scott, S. K. (1983). *Autocatalytic reactions in the isothermal, continuous stirred tank reactor: Isolates, patches, mushrooms, period doubling and chaos*. Chemical Engineering Science, 38(1), 29-43.
- [3] Pearson, J. E. (1993). *Complex patterns in a simple system*. Science, 261(5118), 189-192.